

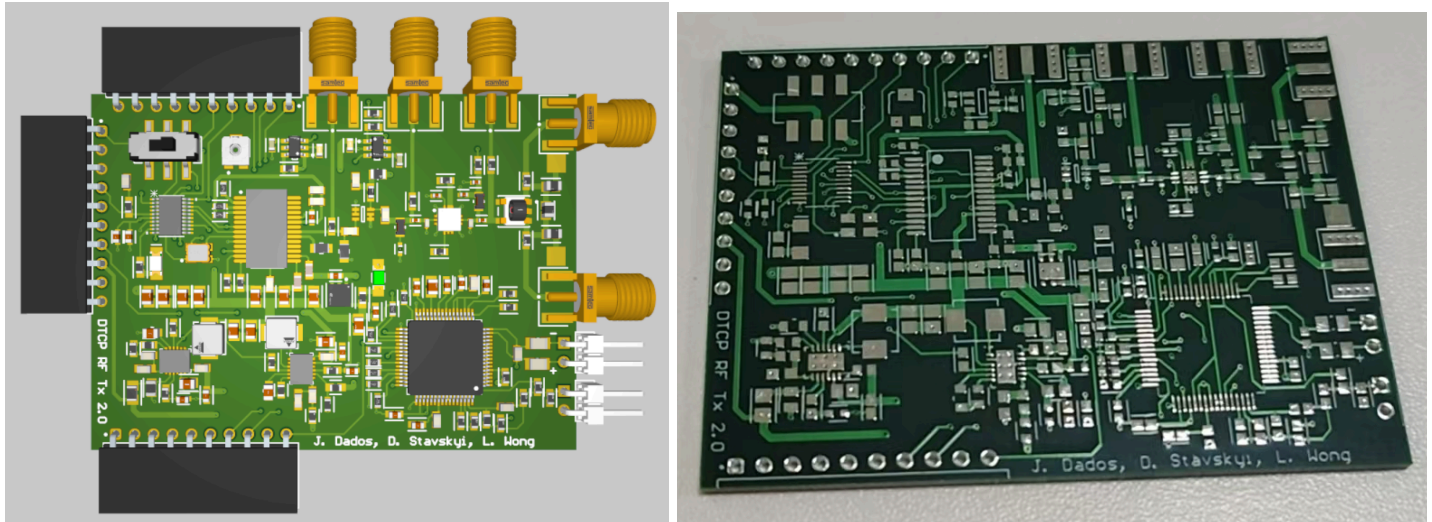
Release Candidate

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Project repository: <https://github.com/jdados/dtcp>

1. Current state of the project work

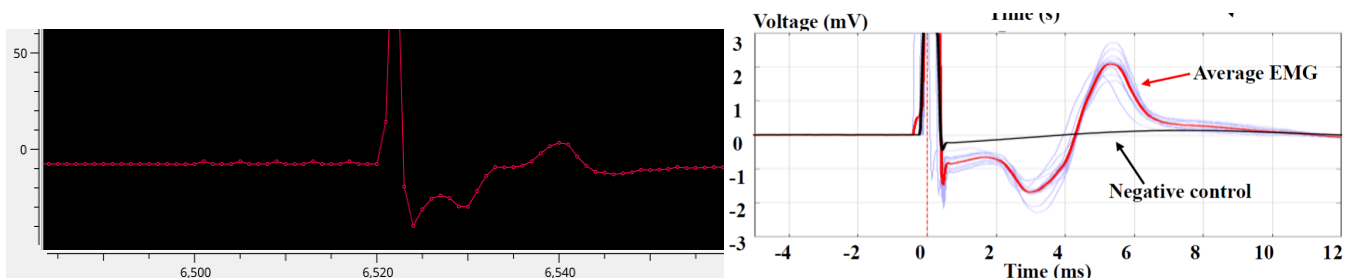
Since the beta build, we designed the final PCB design of the device, integrating the learnings and bug fixes into a single board, implementing all the features of the system:



PCB design in Altium and the fabricated board (photo provided by the manufacturer).

The board is already fabricated and is currently in the shipping process. It is expected to arrive on 3/31 and will be assembled and tested within a week of arrival.

The EMG real-time recording system was expanded by developing a UART-based live plotting system using the Launchpad as a UART-to-USB converter and SerialPlot software as the data capture tool. The functionality was validated by testing it with an artificial signal in saline mimicking the recordings captured by our PhD mentor in live rat recordings (including a large stimulation artifact before the actual EMG signal):



EMG signal captured by our device (on the left) compared to a signal captured by a benchtop system currently used in its place at the Neural Interface Technology Lab.

The sampling rate of the EMG recording circuit was increased to 8 kS/s to allow for the capture of the brief voltage changes related to muscle contraction that follow the stimulation artifact. The EMG recording will be validated in a live rat model in the first two weeks of April.

2. Usability

a. Interface

In addition to all the interfaces developed in the Alpha and Beta builds, we added a UART plotting interface accessed through the SerialPlot app on PC. It allows the user to view a real-time, high sampling rate plot of the EMG waveform.

b. Navigation and perception

After expanding the EMG recording functionalities, the controls required are the power supply/battery connections, the output power setting potentiometer, and a USB connection to a PC running SerialPlot.

c. Responsiveness

The power switch and power potentiometer were replaced with smaller, surface-mount versions that enable a smaller board profile while keeping the same level of user responsiveness.

3. Build quality

a. Robustness and Consistency

Since the beta build, all major functionalities have been fully implemented and validated to ensure no bugs. The board operates bug-free over tests spanning several hours.

b. Aesthetic rigor

The board layout was made following the aesthetics of production-level PCBs. That involved ensuring short signal paths and logical placement of the components on the board.

4. Vertical features

a. External interface

All the external interfaces are verified and carry out functionality correctly. The only minor bug present is the need to re-synchronize the UART datastream several times at the beginning of the connection.

b. Persistent State

The UART interface enables persistent data storage through the SerialPlot functionality of saving a CSV file of the data stream. Hence, the only persistent state of interest in the project has been implemented successfully.

c. Internal Systems

The PCB design of the Release Candidate captures all learnings and bug fixes from the beta version. The communications between all internal systems (such as SPI between MCU and AFE) have been redesigned in hardware to eliminate any issues.